

# *Markov Point Process for Multiple Object Detection*

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**07/03/2023**

# Outline

Motivation

The different issues :

- objects
- reference measure
- prior
- data term
- **optimization**

Some results



## Du contexte



## A la géométrie



Du contexte



A la géométrie



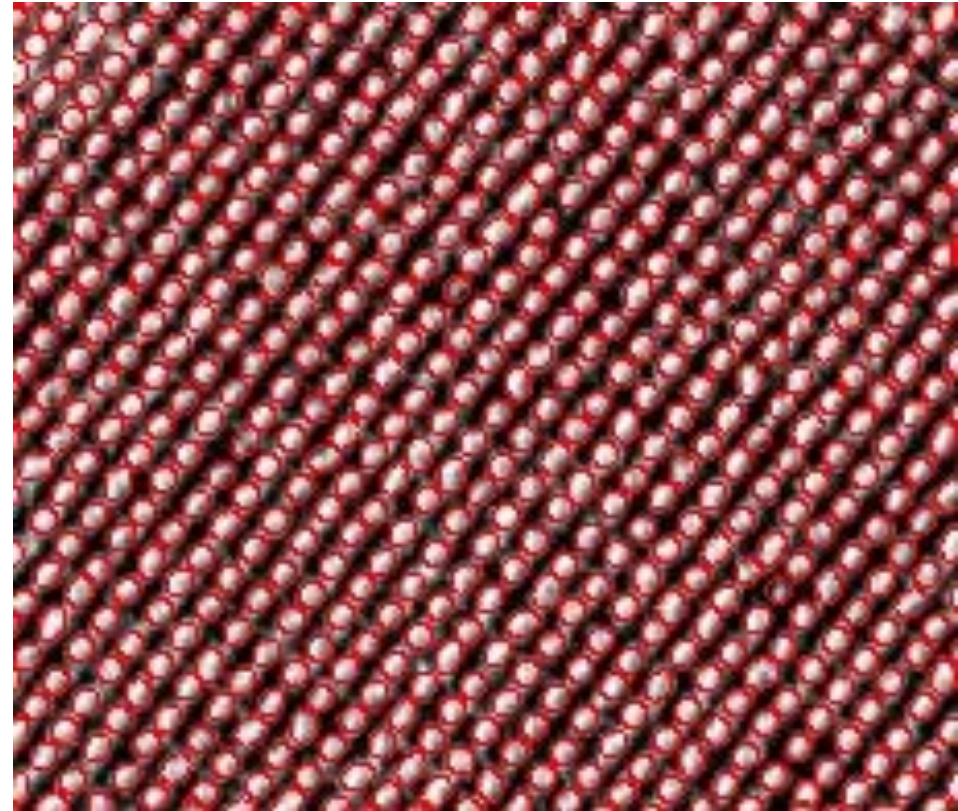
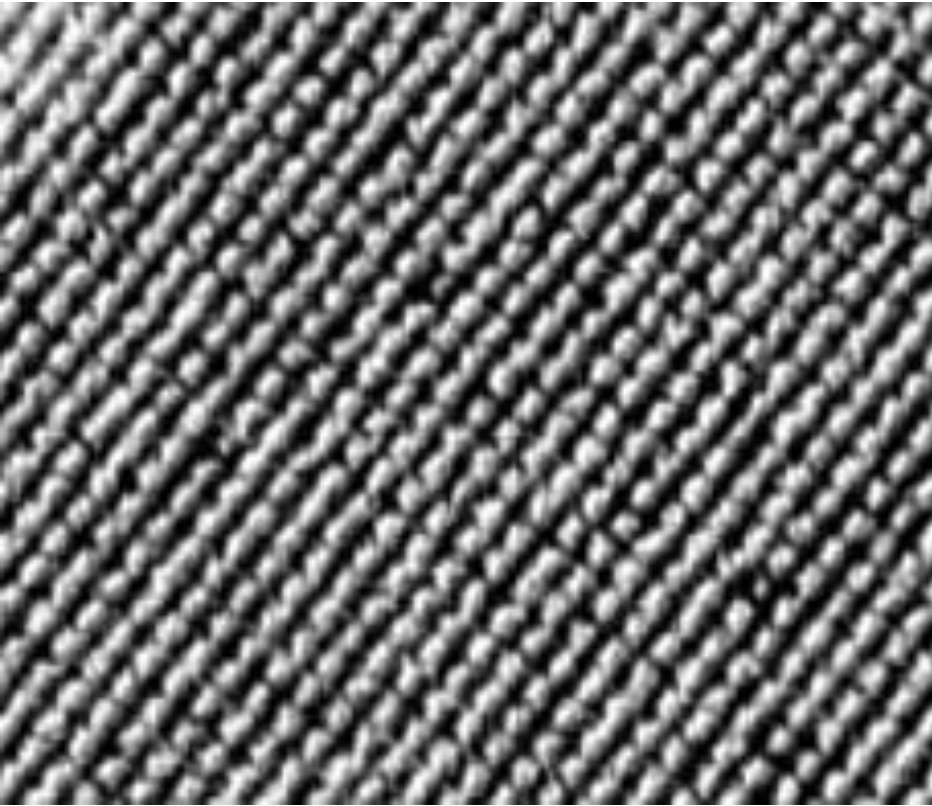
# Motivation : from context to geometry

- 1) Consider prior information (Bayesian approach, Markov Random Fields, interactions)
- 2) Embed geometric information (graph of objects)
- 3 ) Modeling the scene structure (interactions between objects, unknown number of objects)
- 4) Need algorithms for simulating, optimizing the models

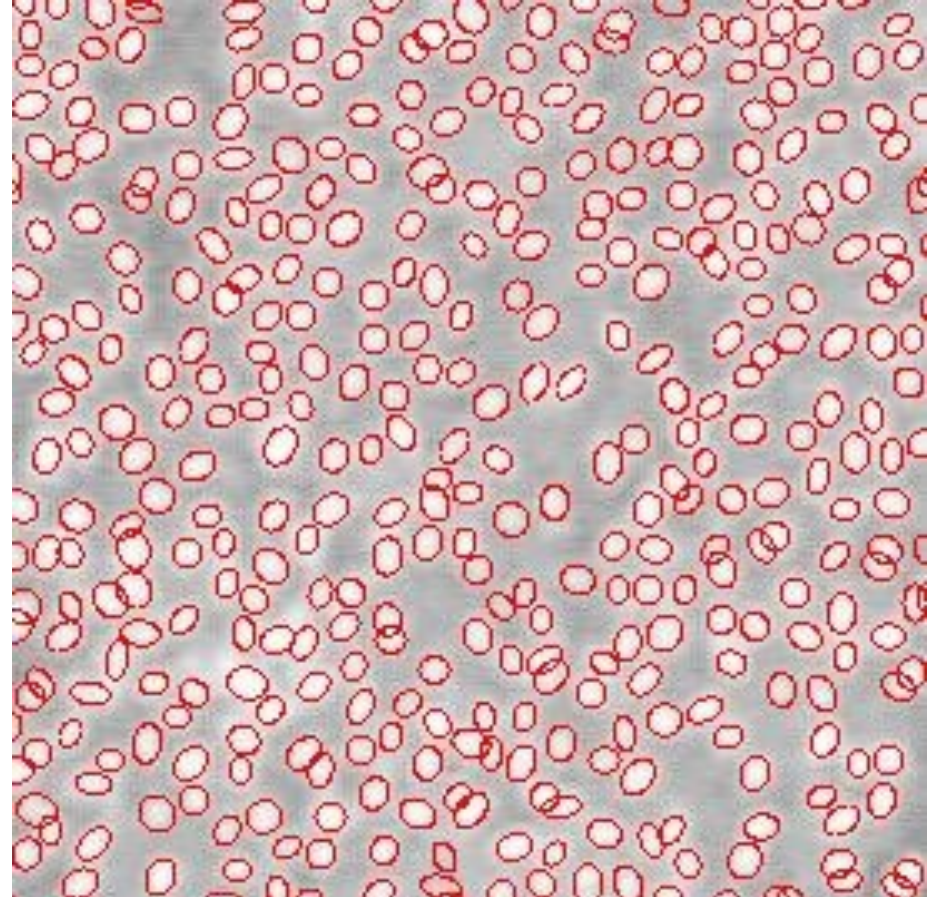
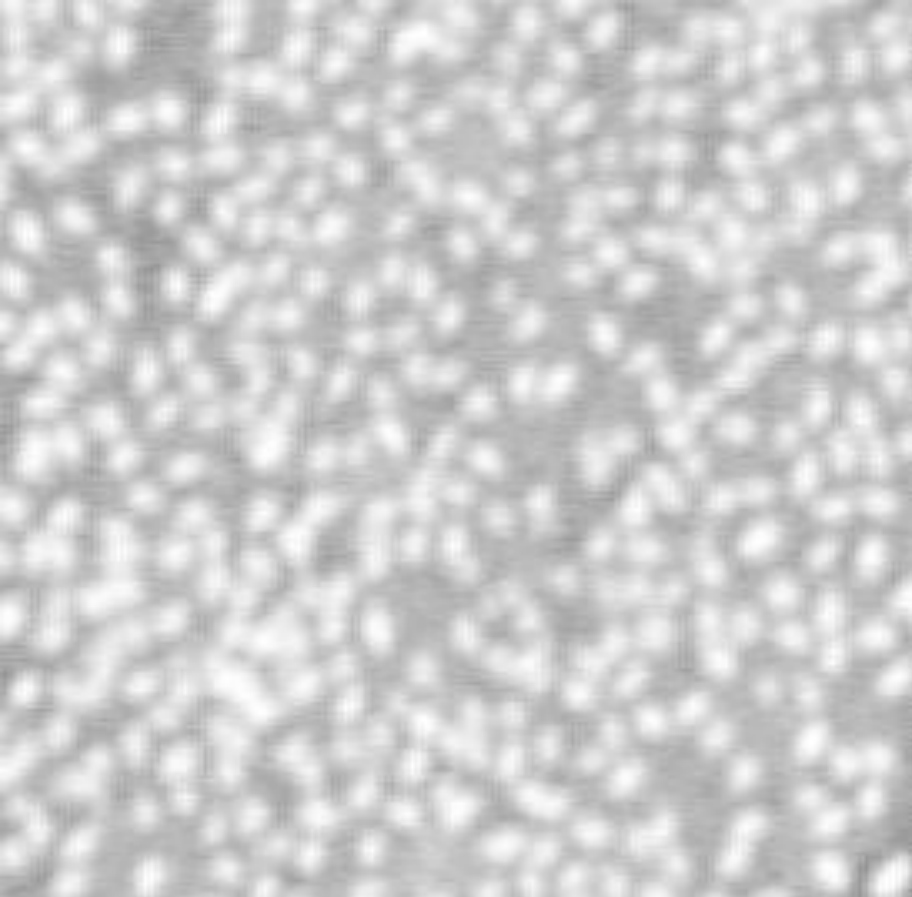


Marked point processes

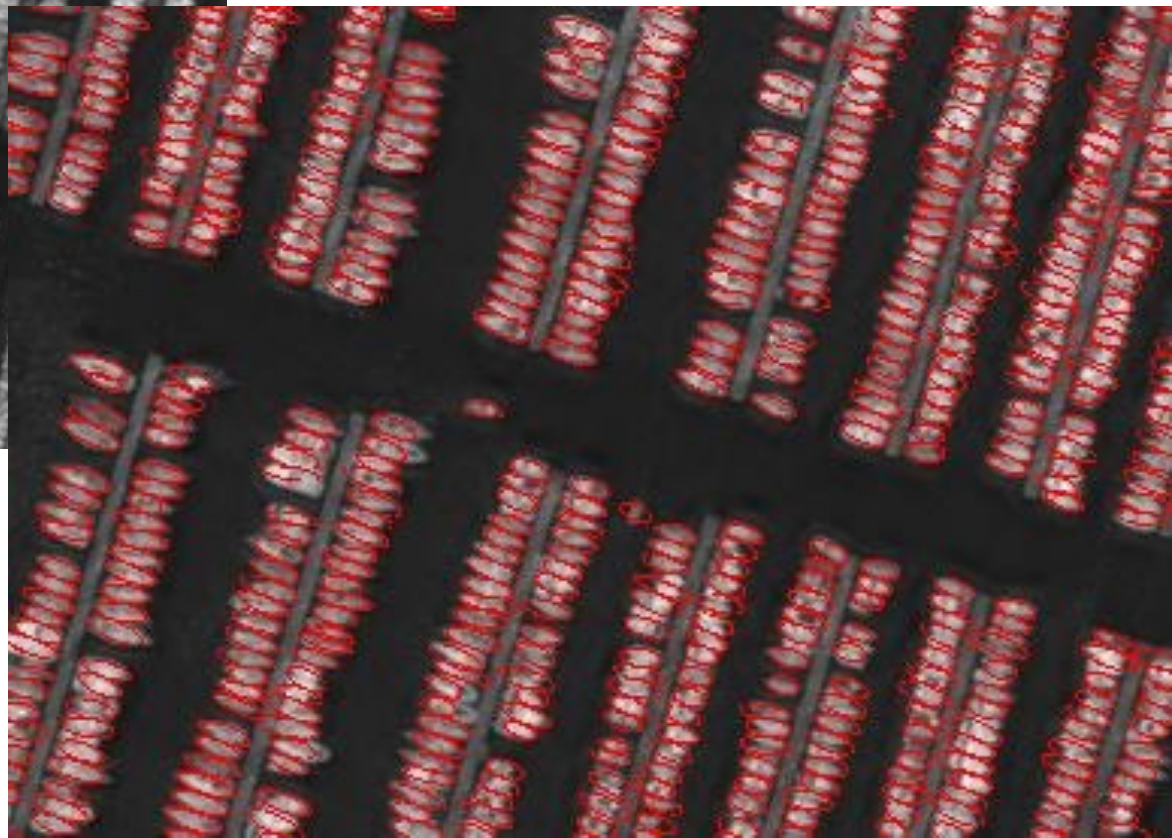
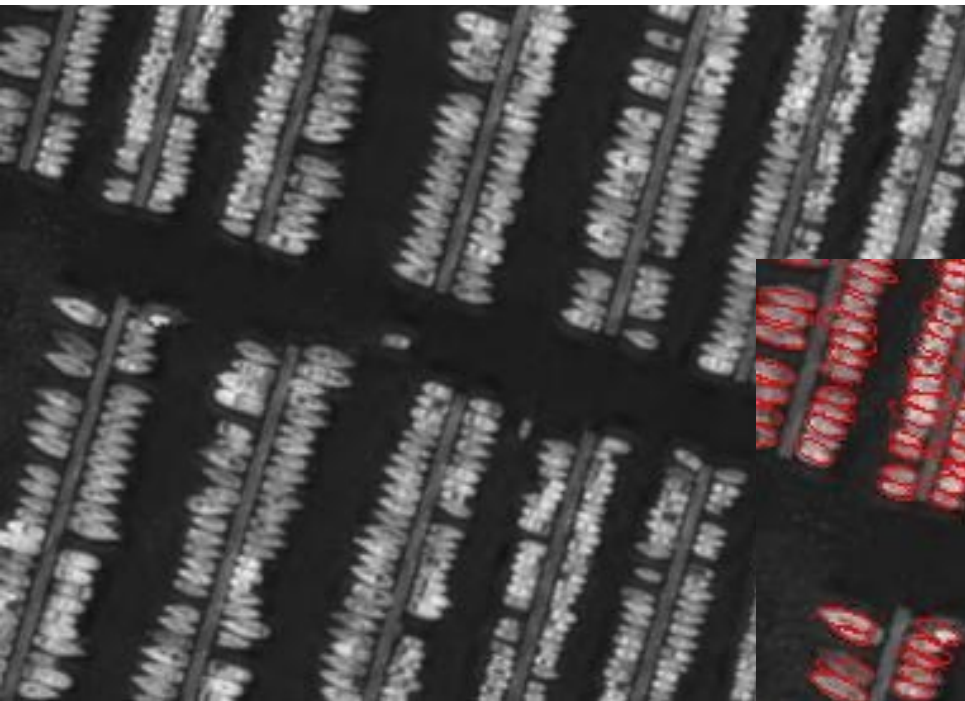
# Motivation : a scene as a collection of objects



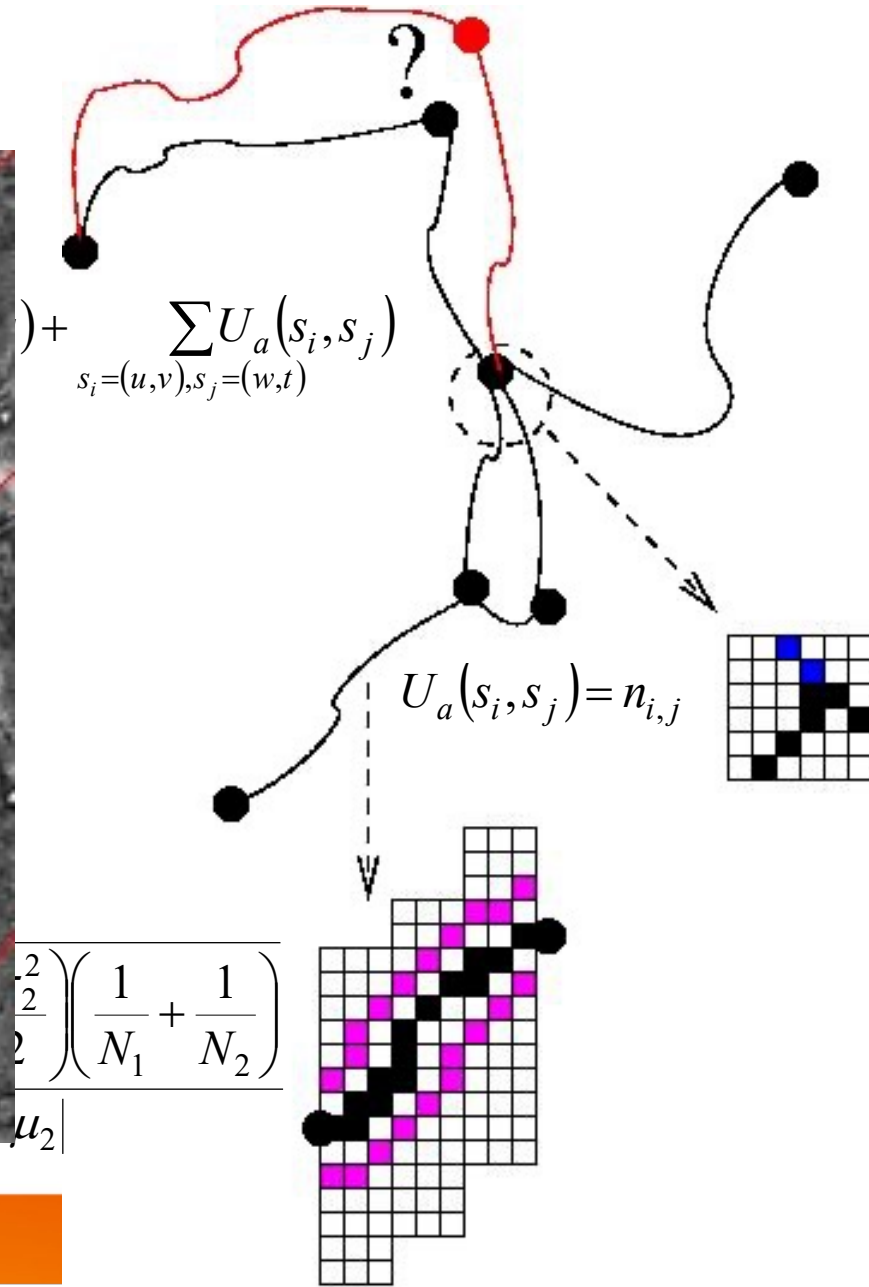
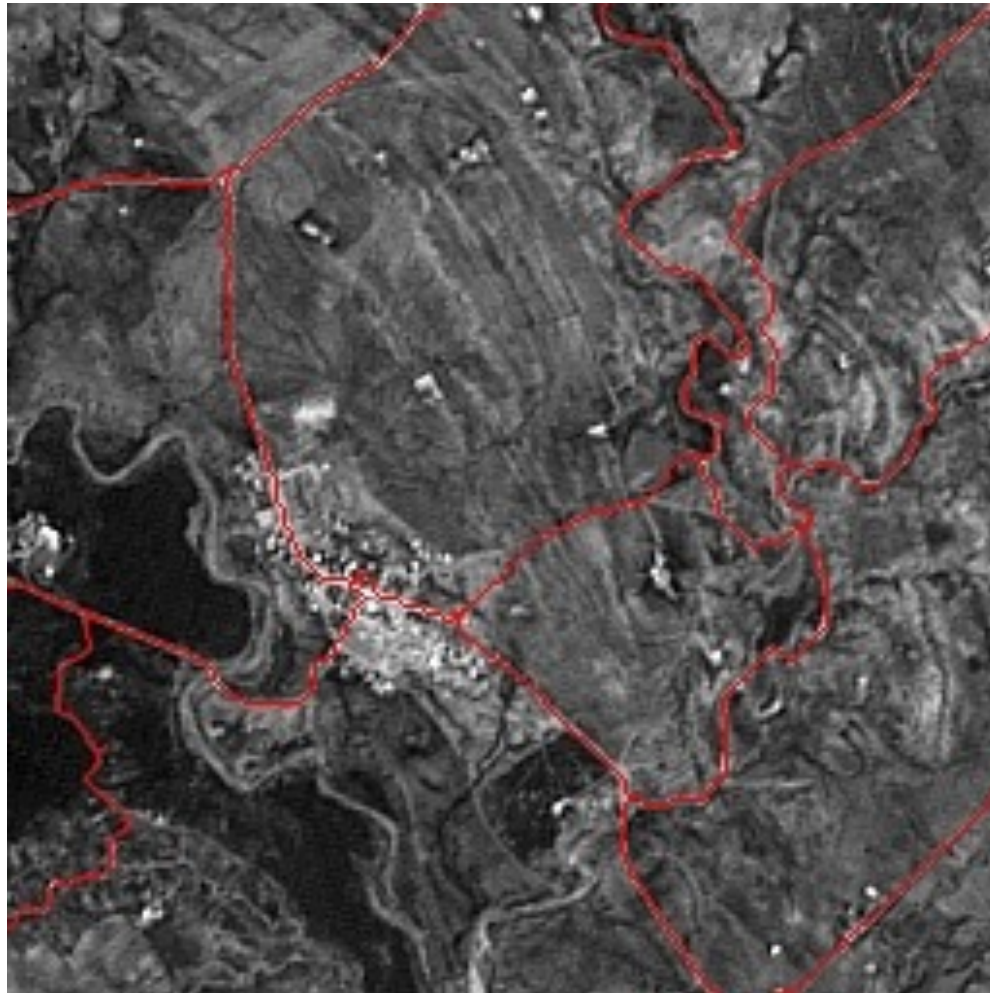
# Motivation : a scene as a collection of objects



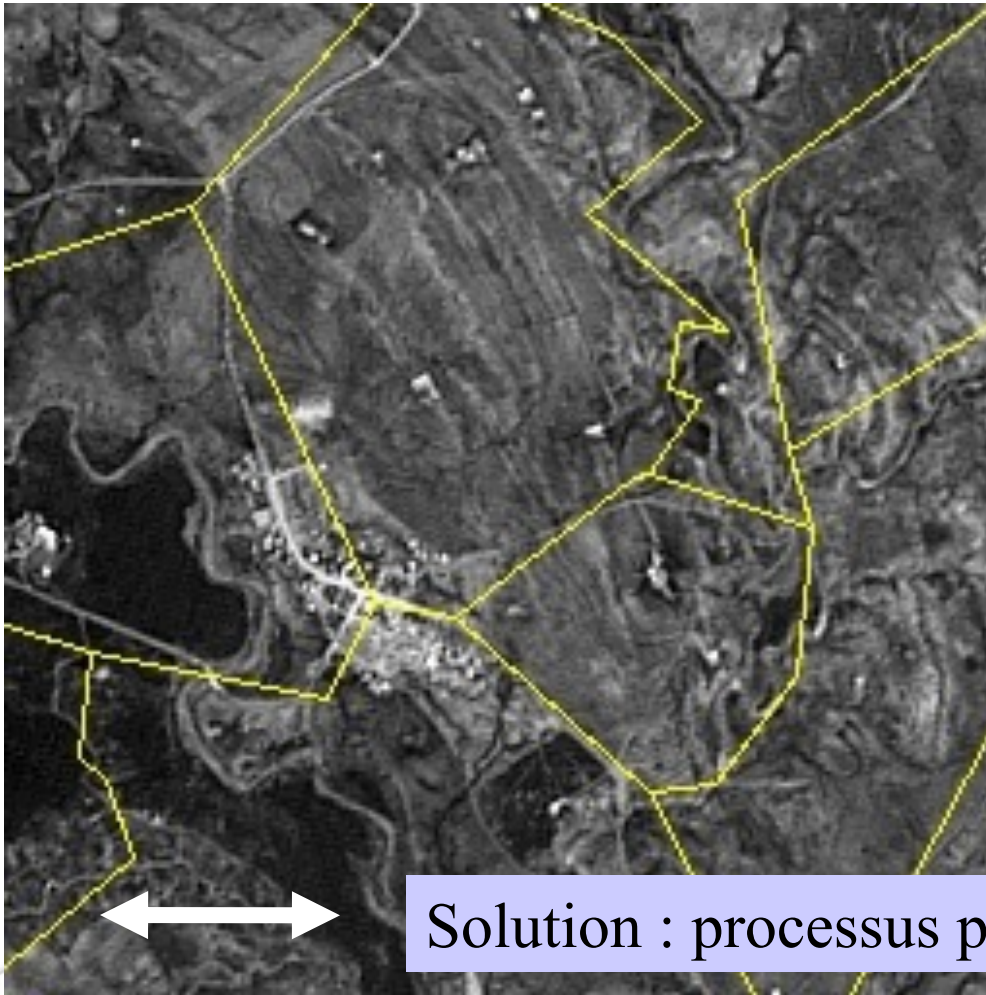
# Motivation : a scene as a collection of objects



# Exemple: détection de réseaux routiers



# Problèmes liés à cette approche



- 1) Combien de noeuds ?
- 2) Où positionner les arêtes ?

Solution : processus ponctuels marqués

# Processus de Poisson

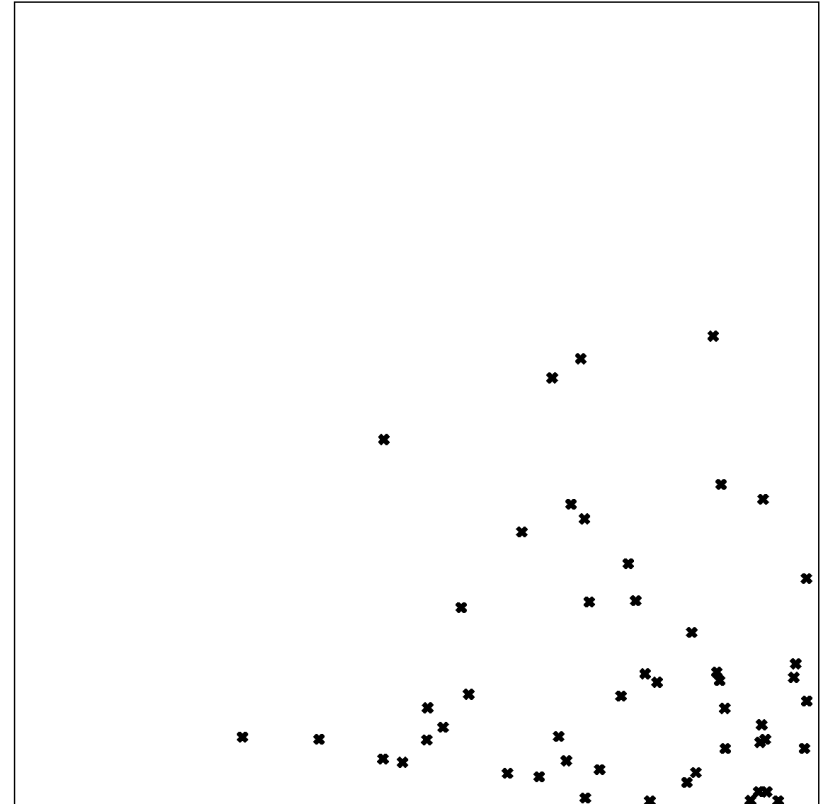
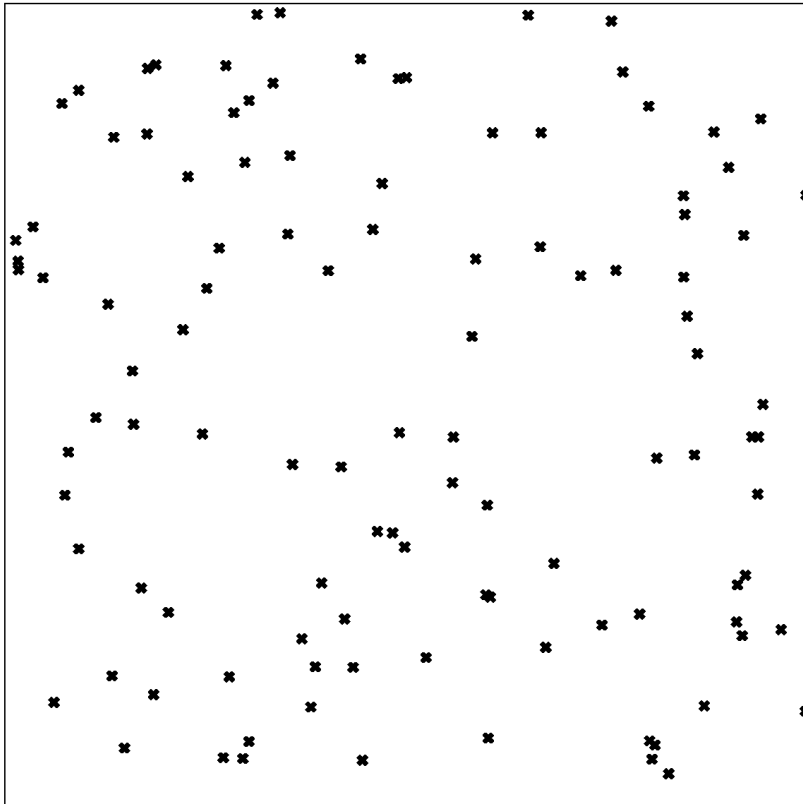
- Un processus ponctuel  $X$  sur  $\chi$  est appelé processus ponctuel de Poisson de mesure d'intensité  $\nu(\cdot)$  si :
  - $N_X(A)$  suit une loi de Poisson d'espérance  $\nu(A)$  pour tout borélien borné  $A$  de  $\chi$
  - Pour  $k$  boréliens disjoints  $A_1, \dots, A_k$ , les variables aléatoires  $N_X(A_1), \dots, N_X(A_k)$  sont indépendantes.

# Intensité

- Processus de Poisson homogène :  $\nu(\cdot)$  proportionnelle à la mesure de Lebesgue  $\Lambda(\cdot)$
- Processus non homogène : fonction d'intensité  $\lambda(\cdot) > 0$  définie comme la dérivée de Radon Nikodym de  $\nu(\cdot)$  par rapport à la mesure de Lebesgue :

$$\nu(A) = \int_A \lambda(x) \Lambda(dx) < \infty$$

# Simulations du processus de Poisson



Homogène vs non homogène

# Processus ponctuels définis par une densité

- On définit  $X$  par sa densité de probabilité  $f$  (dérivée de Radon Nikodym) par rapport à la loi  $\pi_v(.)$  du processus de Poisson dit de référence :

$$f : N^f \rightarrow [0, \infty[ : \int_{N^f} f(x) d\pi_v(x) = 1$$

- On a alors :

$$p_n = \frac{e^{-v(\chi)}}{n!} \int_{\chi} \cdots \int_{\chi} f(\{x_1, \dots, x_n\}) dv(x_1) \cdots dv(x_n)$$

# Processus de Strauss

$$f(\mathbf{x}) = \alpha \beta^{n(\mathbf{x})} \gamma^{s(\mathbf{x})}$$

$n(\mathbf{x})$  : nombre de points de  $\mathbf{x}$

$s(\mathbf{x})$  : nombre de paires de points en relation (nombre de cliques)

$u \sim v$  si et seulement  $d(u,v) < R$

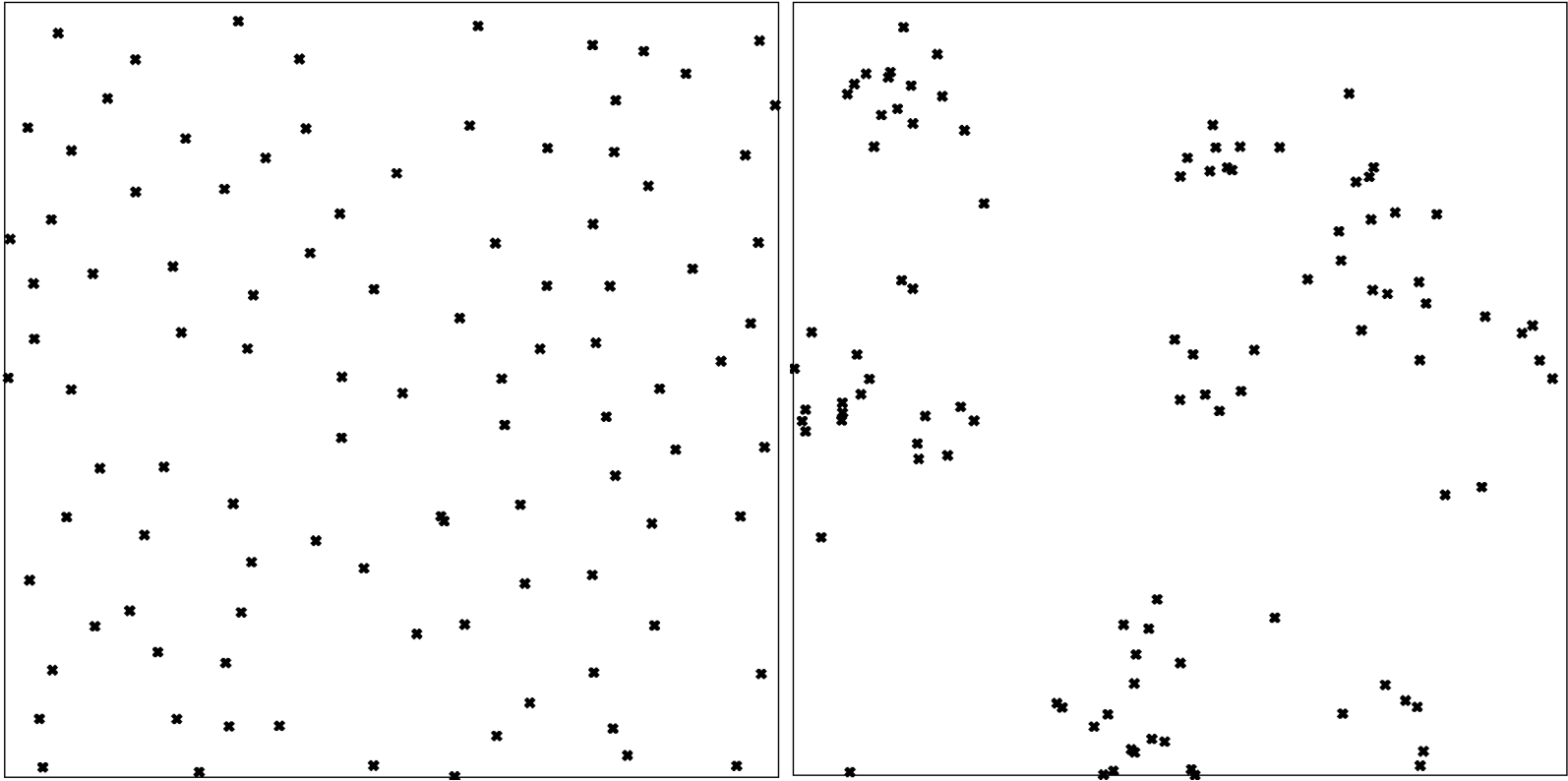
$\gamma = 1$  : processus de Poisson d'intensité  $\beta\lambda(\cdot)$

$0 < \gamma < 1$  : répulsion entre points proches

$\gamma = 0$  : interdiction de points proches (hard core process)

$\gamma > 1$  : processus attractif ou agrégé si l'on fixe le nombre de points

# Simulations du processus de Strauss



Répulsif

vs

Attractif

# The configuration space

« Simple » parametric shapes :  $S = \{s = (x, m), x \in K, m \in M\}$



$(x, r)$



$(x, a, b, \theta)$



$(x, L, l, \theta)$



$(x, L, \theta)$

$$\Omega_n = \{\{s_1, \dots, s_n\}, s_i \in S\}$$

$$\Omega_0 = \emptyset$$

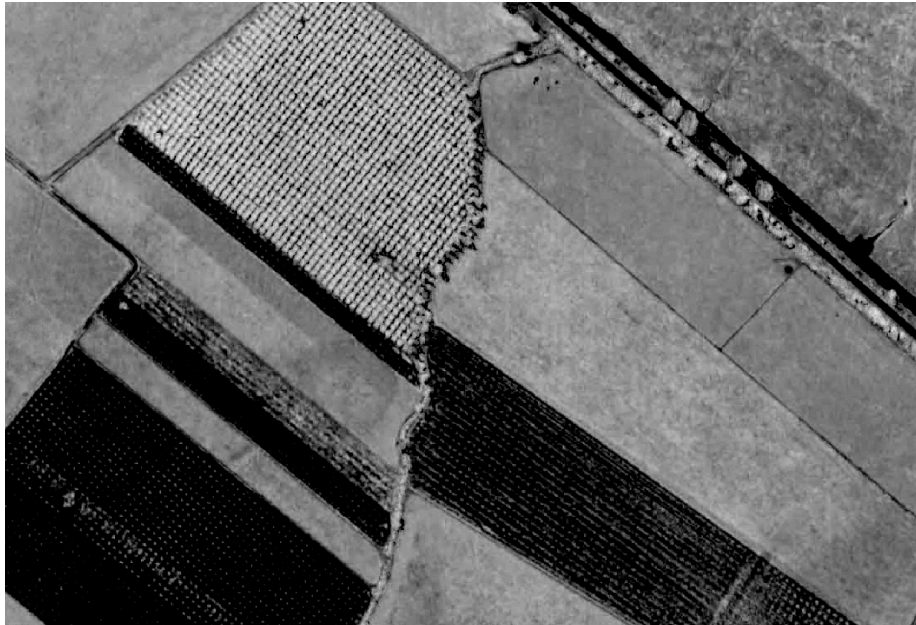
$$\Omega = \bigcup_n \Omega_n$$

# The reference measure

Usually the Poisson measure :

$$\pi_v(B) = e^{-v(\chi)} \left( 1_{[\emptyset \in B]} + \sum_{n=0}^{\infty} \frac{\pi_{v_n}(B)}{n!} \right)$$

$$\pi_{v_n}(B) = \int_v \cdots \int_v 1_{[\{x_1, \dots, x_n\} \in B]} v(dx_1) \cdots v(dx_n)$$



Intensity measure: uniform or not

$$v(A) \int_A \lambda(x) dx$$

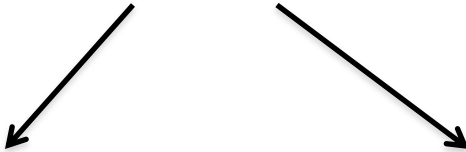
NDVI MAP

# The density

The model is defined by a density (usually un-normalised)  
w.r.t the reference measure:

$$f : \Omega \rightarrow [0, \infty[, \int_{\Omega} f(x) d\pi_v(x) < \infty$$

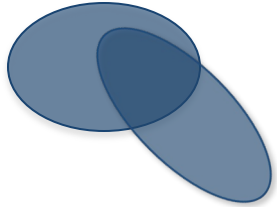
Mimicking the Bayesian approach:

$$f(x) = g(x)h_I(x)$$


Prior                      Data (I) term

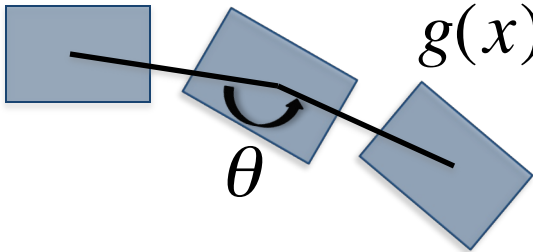
# The prior

Overlap penalization (pairwise interaction):



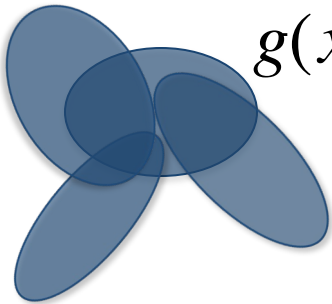
$$g(x) = \prod_{i \sim j} \varphi(x_i, x_j) \quad \varphi(x_i, x_j) = \Phi \left( \frac{|s_i \cap s_j|}{\min(|s_i|, |s_j|)} \right)$$

Alignment :



$$g(x) = \prod_{i \sim j \sim k} \varphi(x_i, x_j, x_k) \quad \varphi(x_i, x_j, x_k) = \Phi(|\pi - \theta_{i,j,k}|)$$

Overlap penalization :



$$g(x) = \prod_i \varphi(x_i, x_j, j \sim i) \quad \varphi(x_i, x_j, j \sim i) = \Phi \left( \max_{j \sim i} \left( \frac{|s_i \cap s_j|}{\min(|s_i|, |s_j|)} \right) \right)$$

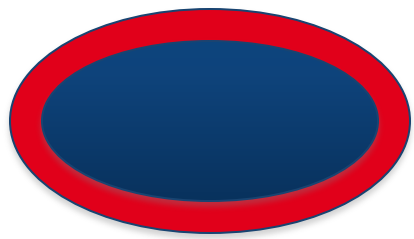
And many more ...

# The data term

Bayesian approach: likelihood

$$h_Y(x) = \prod_{i \text{ inside objects}} 1_{\text{object}}(y_i) \prod_{i \text{ outside objects}} 1_{\text{background}}(y_i)$$

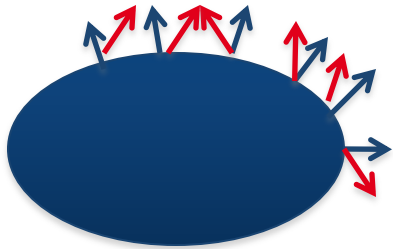
Distance between interior and exterior



$$h_Y(x) = \prod_{\text{objects}} \text{dissimilarity}(\text{red pixels, blue pixels})$$

Examples : Bhattacharya distance, Statistical test,...

Geometrical consistency



$$h_Y(x) = \prod_{\text{objects}} \exp - U(\omega) \quad U(\omega) = \frac{1}{|\partial\omega|} \int \left\langle \frac{\nabla Y(u)}{\sqrt{\|\nabla Y(u)\|^2 + \varepsilon^2}}, n(u) \right\rangle du$$

$$U_d(\omega, Y) = \psi(U(\omega), t)$$

And many more ...

# Optimization : RJMCMC

$$(h(x))^{1/T} d\pi(x)$$

- initialize the temperature  $T$  and the configuration  $x$  (empty set)
- Choose a proposition kernel  $Q_m(x, \cdot)$  with probability  $p_m(x)$ , or let the configuration unchanged probability  $1 - \sum_m p_m(x)$ .
- Sample  $x'$  according to the chosen kernel
- Compute the acceptance ratio :

$$R_m(x, x') = \frac{D_m(x', x)}{D_m(x, x')} = \frac{(h(x'))^{1/T} \pi(dx') Q_m(x', dx)}{(h(x))^{1/T} \pi(dx) Q_m(x, dx')}$$

- With probability  $\alpha = \min(1, R_m)$  set  $x_{t+1} = x'$ , else reject the proposition :  $x_{t+1} = x$ .

Some perturbation kernels (proposal)

Adding an object

Removing an object

Modifying an object (translation, rotation, dilation)

Merging/Splitting objects

# Optimization : RJMCMC

Pros :

- Generality
- Choice for kernels
- Convergence to the global optimum

Cons :

- Rejection
- Simulated annealing scheme (parameters setting)
- Kernels usually involve one or two objects

# Optimization : Multiple births and deaths

Goals :

Avoid rejection

Consider several objects at once

Idea :

Extend Langevin' s dynamics (Stochastic Differential Equation : diffusion process)

# Optimization : Multiple births and deaths

Consider :  $f(x) = \exp - \beta E(x)$

A birth intensity consisting in adding an object  $u$  to the configuration  $x$  :

$$b(x, u) du = z du \text{ if } u \in D(x), \text{ where } D(x) = \left( \bigcup_{v \in x} B_v(\varepsilon) \right) \cap K$$

A death intensity consisting in removing an object  $u$  from the configuration  $x$  :

$$d(x, u) = \exp \beta [ E(x) - E(x/u) ] \text{ if } u \in x,$$



Detailed balance condition holds

# Optimization : Multiple births and deaths

## A New Approximation Process

Markov Chain :  $T_{\beta, \delta}(m)$ ,  $m = \left\lfloor \frac{t}{\delta} \right\rfloor = 0, 1, 2, \dots$  (discretization of time)

$$x_{n+1} = x_1 \cup x_2, \quad x_1 \subseteq x_n, \quad x_1 \cap x_2 = \emptyset$$

$x_2$  : Poisson law (with intensity  $z$ ) distributed

Birth transition :

$$q_{u, \delta} = \begin{cases} z\delta\Delta u, & \text{if } x \rightarrow x \cup \{u\} \\ 1 - z\delta\Delta u, & \text{if } x \rightarrow x \text{ (no birth in } \Delta y) \end{cases}$$

Death transition :

$$p_{u, \delta} = \begin{cases} \frac{\delta \exp[E(x) - E(x/u)]}{1 + \delta \exp[E(x) - E(x/u)]}, & \text{if } x \rightarrow x/u \\ \frac{1}{1 + \delta \exp[E(x) - E(x/u)]}, & \text{if } x \rightarrow x \text{ (} x \text{ survives)} \end{cases}$$

# Optimization : Multiple births and deaths

## A New Algorithm

1) Precomputing of the data term / birth map

2) Repeat :

2.1) Birth:

For each pixel, add an object with probability :

$$\delta B(E_d(u))$$

2.2) Sort the objects with respect to their data term value

2.3) Death:

For each object  $u$  taken in the list order, remove it with probability :

$$\frac{\delta \exp[\beta(E(x) - E(x/u))]}{1 + \delta \exp[\beta(E(x) - E(x/u))]}$$

# Optimization : Multiple births and deaths

**Theorem:** When  $N \rightarrow \infty$ ,  $\beta \rightarrow \infty$ ,  $\delta \rightarrow 0$ , convergence toward the configuration minimizing the energy

Pros :

- No rejection in the birth step
- Birth does not depends on the temperature
- Convergence to the global optimum

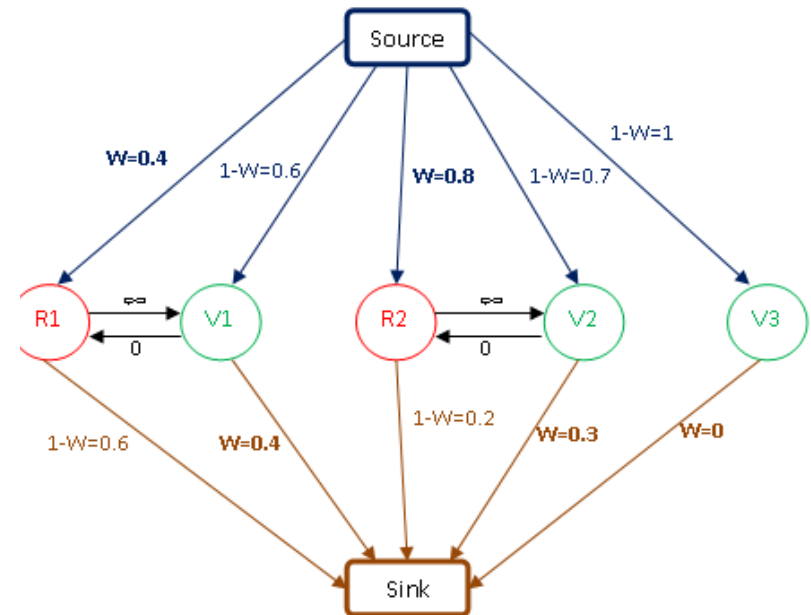
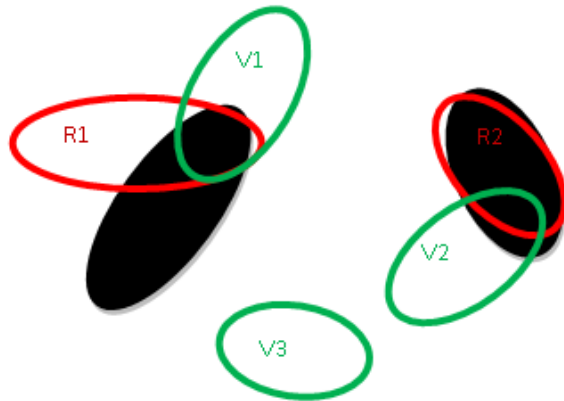
Cons :

- Only births and deaths kernels

# Optimization : Multiple births and cut

Idea : Combine multiples births and deaths with graphcut techniques

- **Generate a first configuration  $x_0$  of non overlapping objects and iterate the following steps:**
- **Birth:**
  - *generate a new configuration of non overlapping objects  $x'$*
- **Death:**
  - *$x_{n+1}$  is defined by  $\text{Cut}(x_n U x')$*



# Optimization : Multiple births and cut

Pros :

- No rejection in the birth step
- No cooling schedule

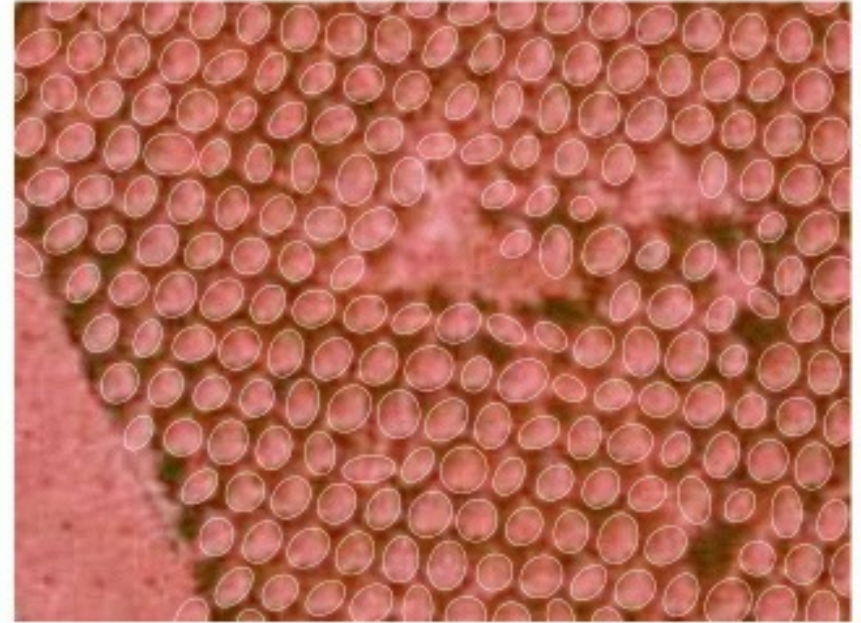
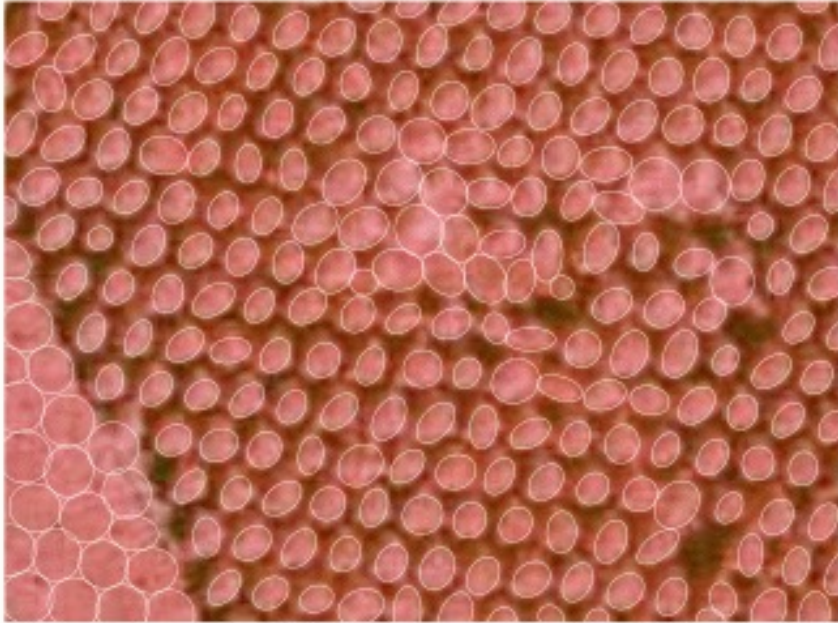
Cons :

- Only births and deaths
- No proof of convergence

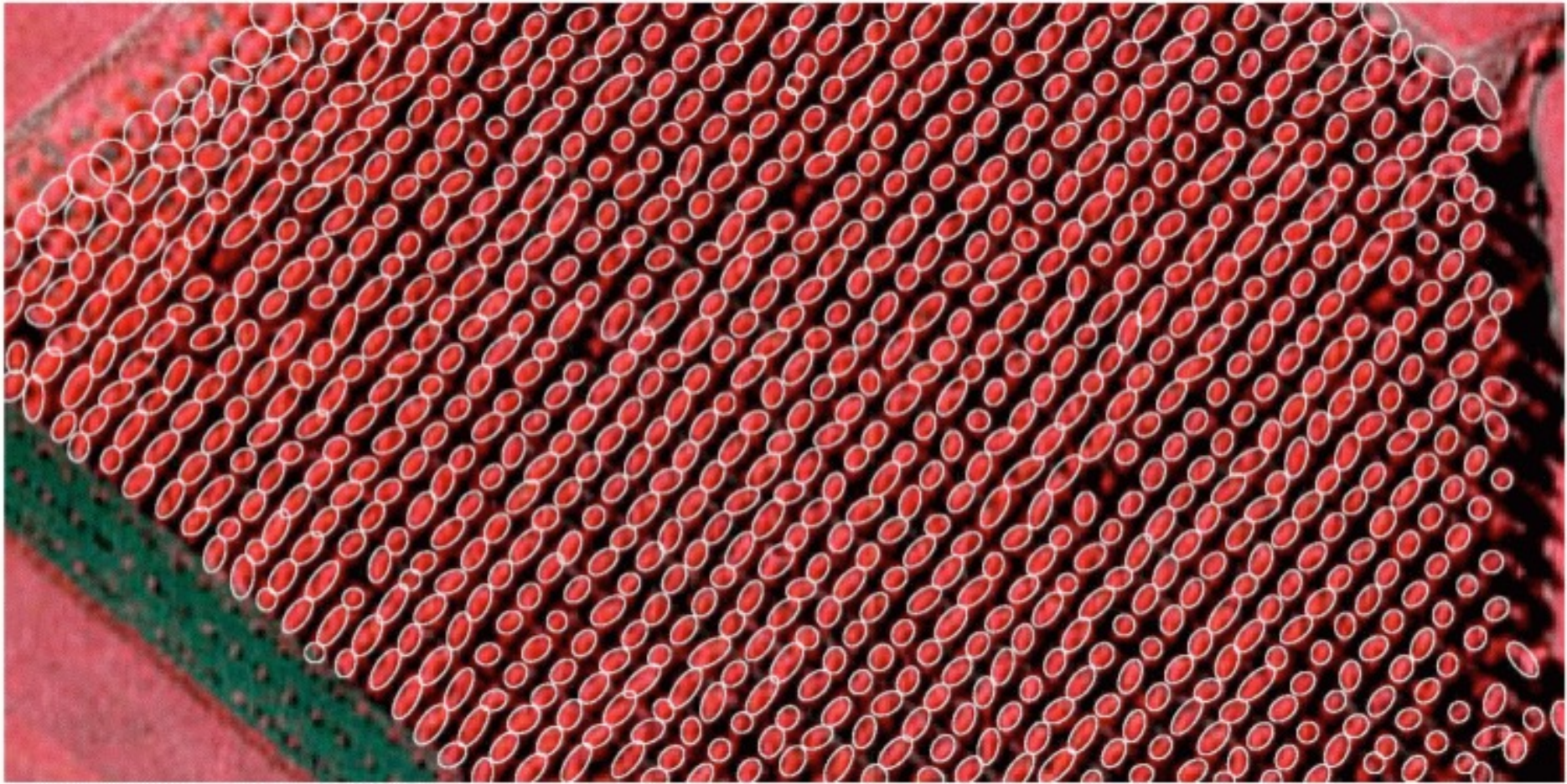
# Result : Trees counting



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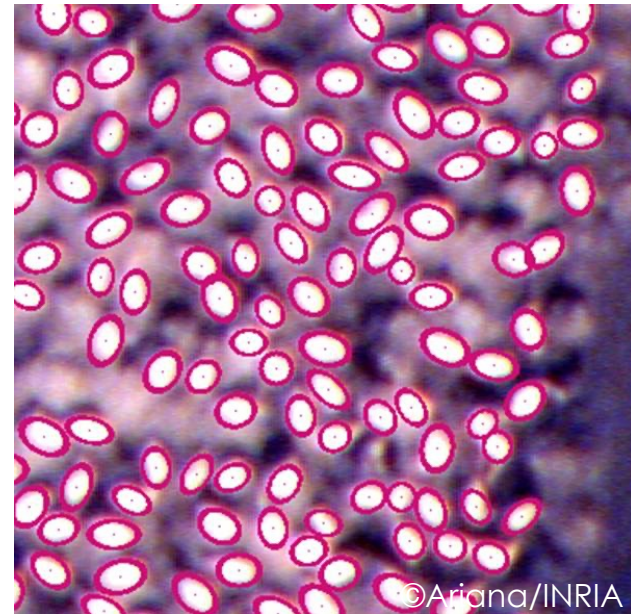
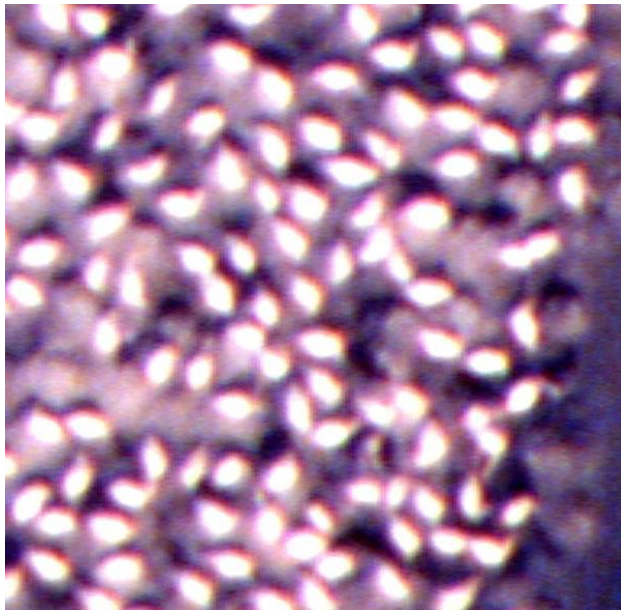
# Result :Trees counting



# Result : Flamingos counting

Estimation of the size of a colony in Turkey (2004):

♣ 3682 detected flamingos (*Tour du Valat*: 3684 flamingos)



# Result : Flamingos counting



## Estimation of the size of a colony in Mauritania:

♣ Very high density



# Result : Flamingos counting



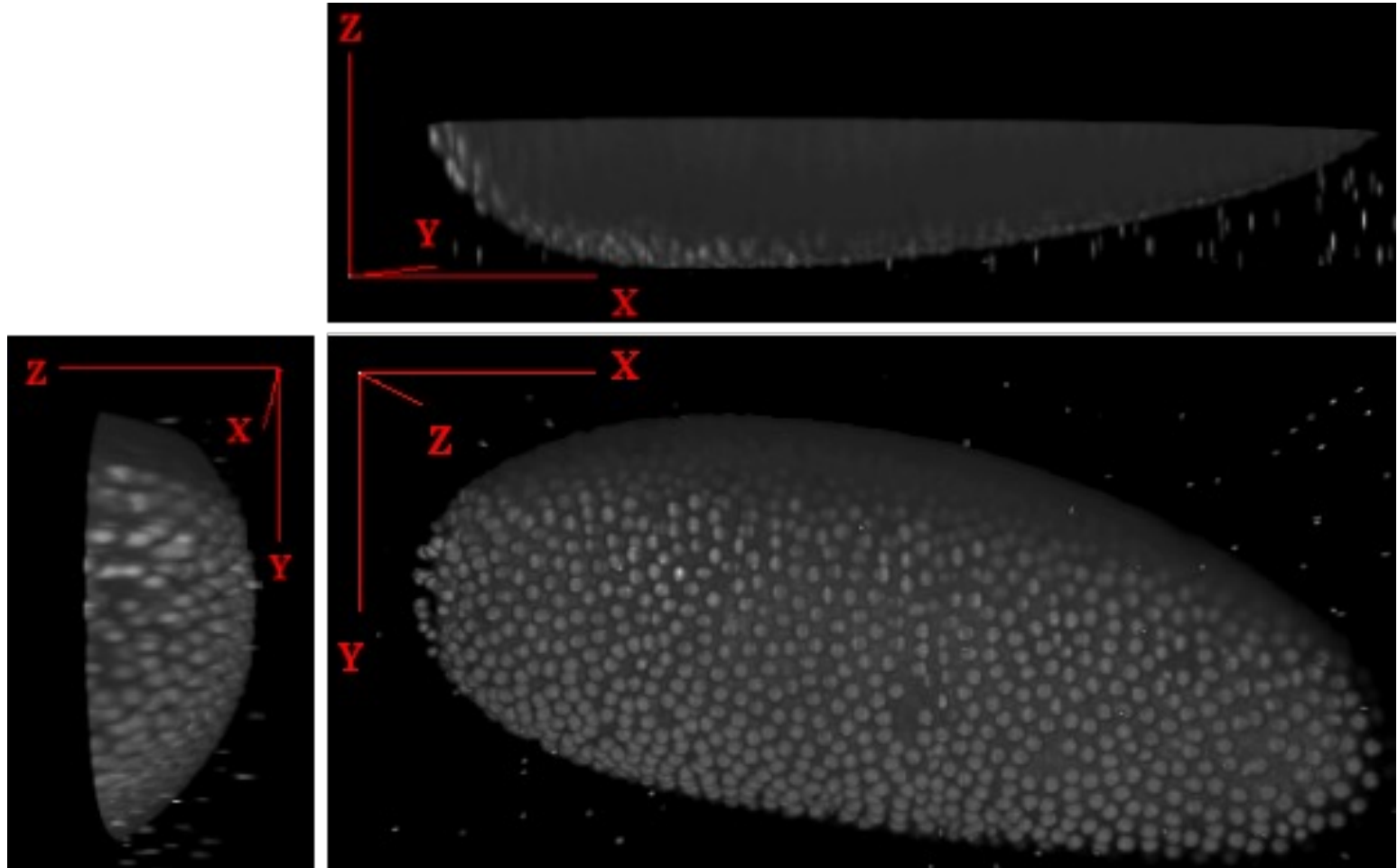
## Estimation of the size of a colony in Mauritania:

♣ 14595 detected flamingos (*Tour du Valat: 13650 flamingos*)

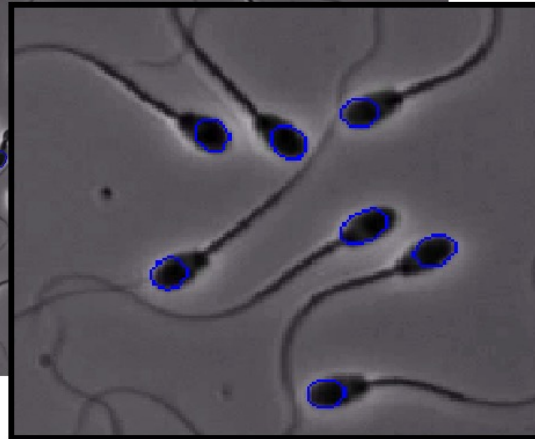
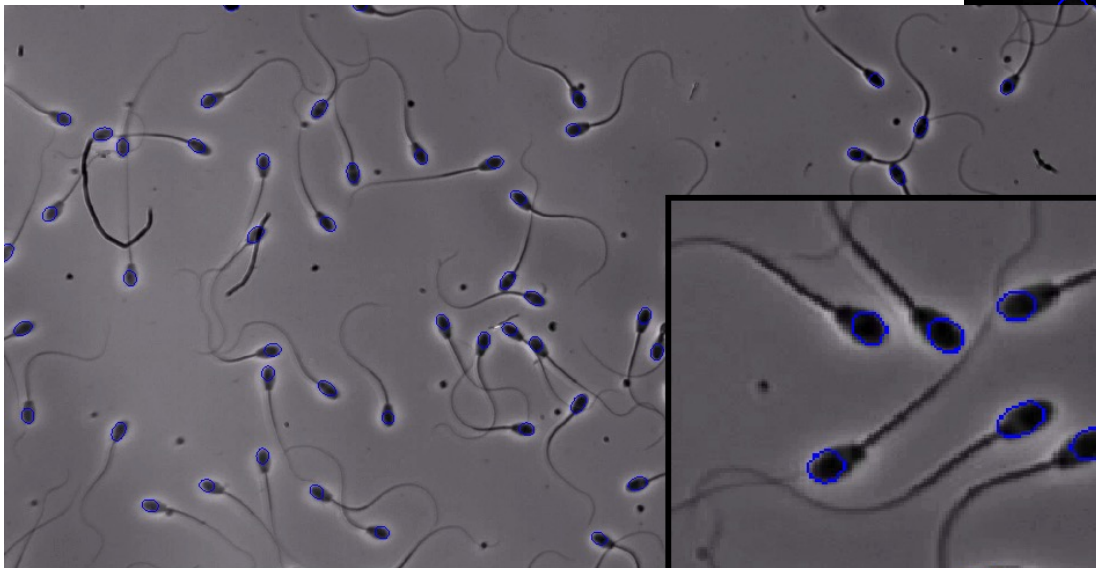
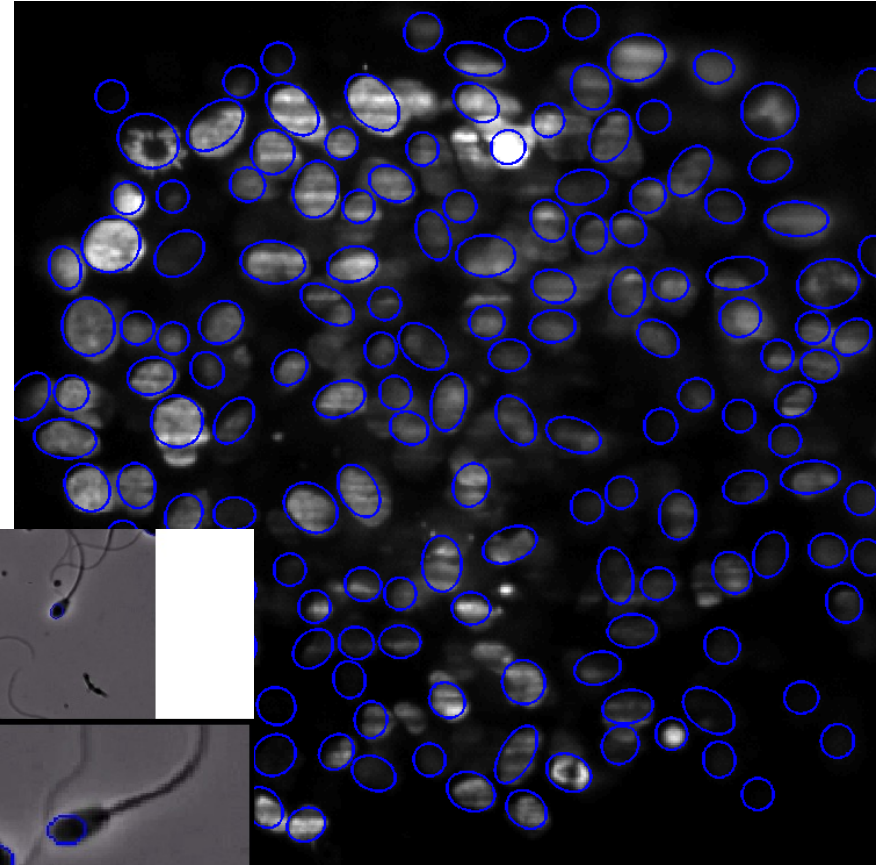
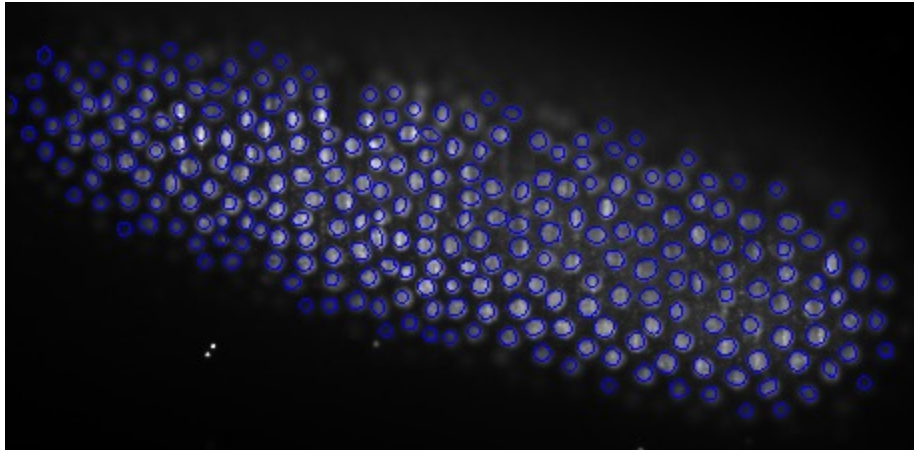


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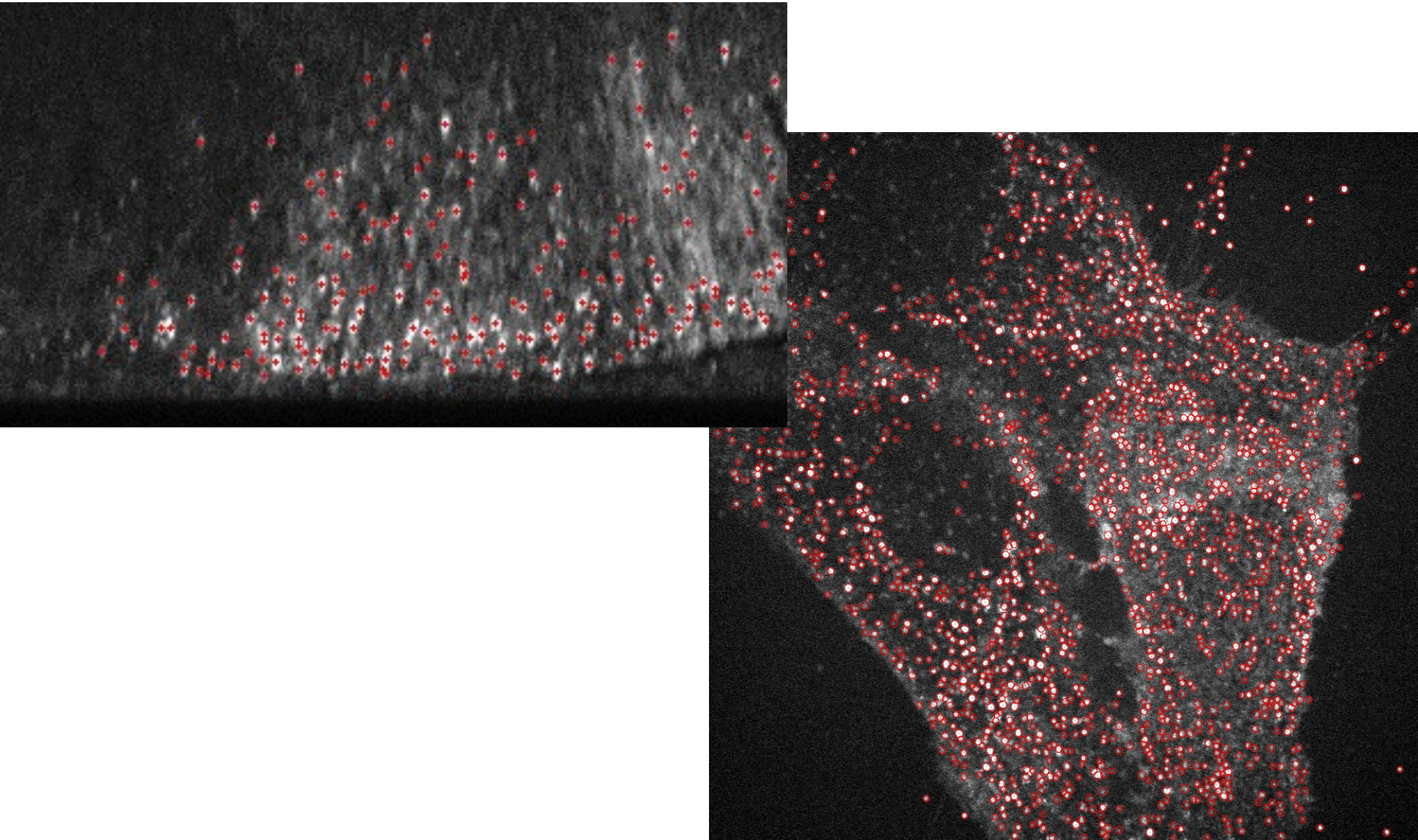
# Result : Cells counting



# Result : Cells counting



# Result : Vesicle (co)localization



# Conclusion

Pros :

General framework / Numerous application  
Embed strong geometric constraint

Future work :

parallelism  
parameter estimation  
open source software